GENO-MAP: Correspondence-Free Diagnostics for Sweet Potato Diversity Maps

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Correspondence-free validation of disjoint DArT panels via geometry diagnostics and robustness curves; PCA is the stable analytic baseline in $n \ll p$.

1. Why Correspondence-Free?

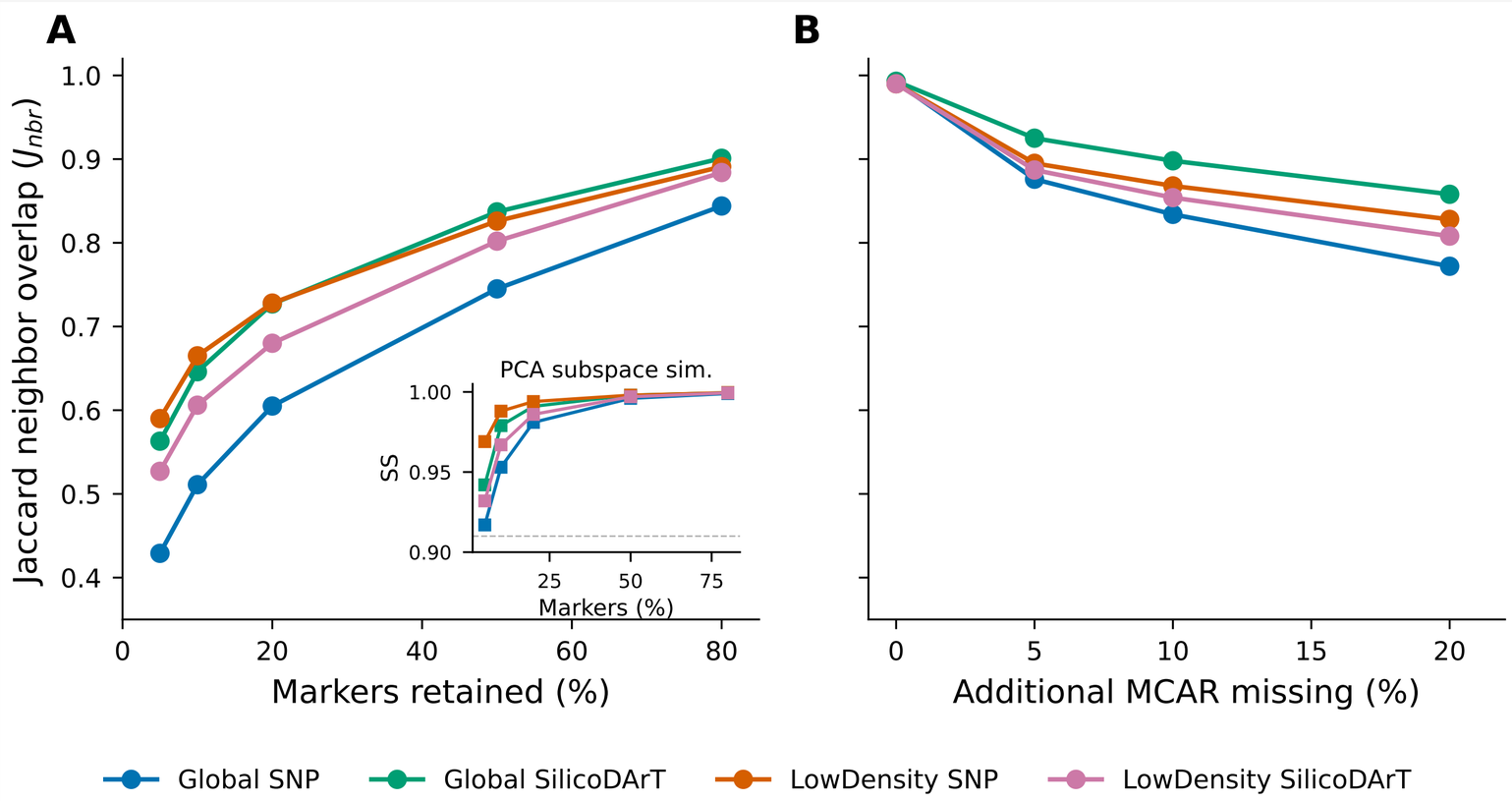
Sweet potato DArT panels use **disjoint ID namespaces** (CIP accessions vs. plate/well coordinates). Without an external manifest, cross-panel alignment is infeasible. We validate each panel *independently* via geometry diagnostics and robustness curves.

Panel	n	p	n/p
G-SNP	5 970	20 069	0.30
G-Silico	5 970	57 715	0.10
LD-SNP	630	62 732	0.01
LD-Silico	635	38 272	0.02

2. Method (fixed graph protocol)

- **PCA-30D** as deterministic analytic space (graph built here)
- **kNN graph**: $k = 15$, cosine distance on PCA scores
- **UMAP-2D** for visualization only (stochastic)
- **Validation**: geometry diagnostics + robustness perturbations (marker dropout 5–80%, MCAR +0–20%)

3. Robustness Under Perturbation

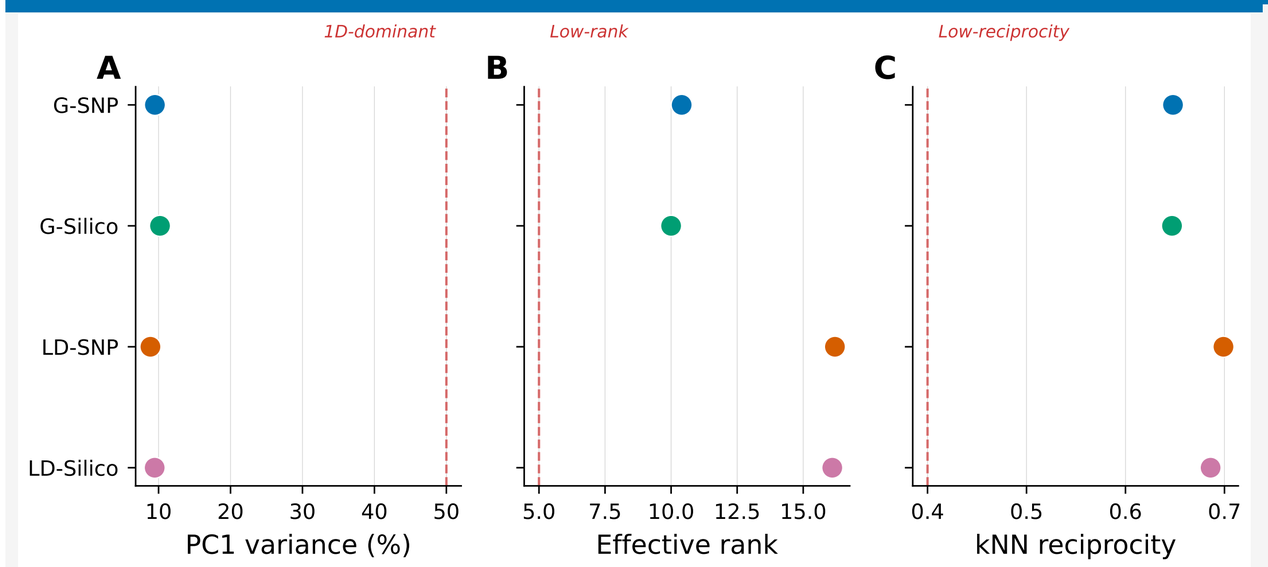


(A) Marker subsampling (5–80% retention): kNN neighborhood overlap (J_{nbr}) degrades smoothly from 0.90 to 0.43 as markers are removed. The PCA subspace similarity remains ≥ 0.91 even at 5% retention, confirming that the principal geometry is genome-wide, not marker-specific.

(B) MCAR injection (+0–20%): J_{nbr} drops to 0.77–0.86 at +20% additional missing values. The PCA subspace is practically invariant (SS $>$ 0.998).

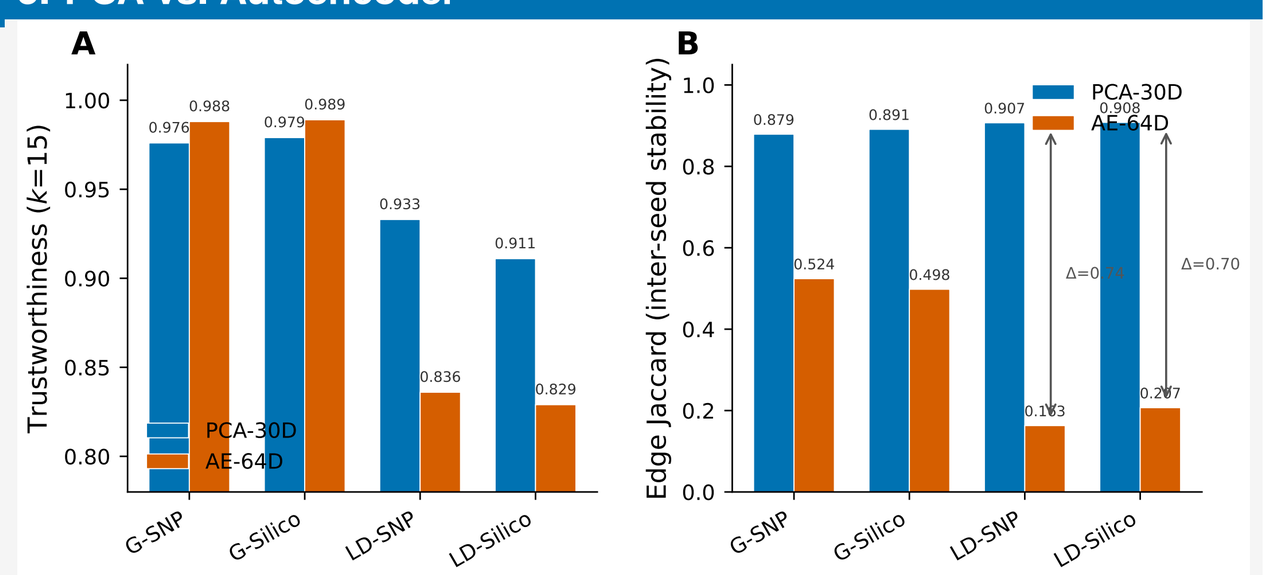
Degradation is **smooth and monotonic** across all four panels — **no cliff effects**, enabling data-quality thresholds for germplasm curation workflows.

4. Panel Geometry & QA



QA thresholds (dashed): PC1 $>$ 50%, eff. rank $<$ 5, reciprocity $<$ 0.40. All panels show **no 1D collapse** and **strong reciprocity**.

5. PCA vs. Autoencoder

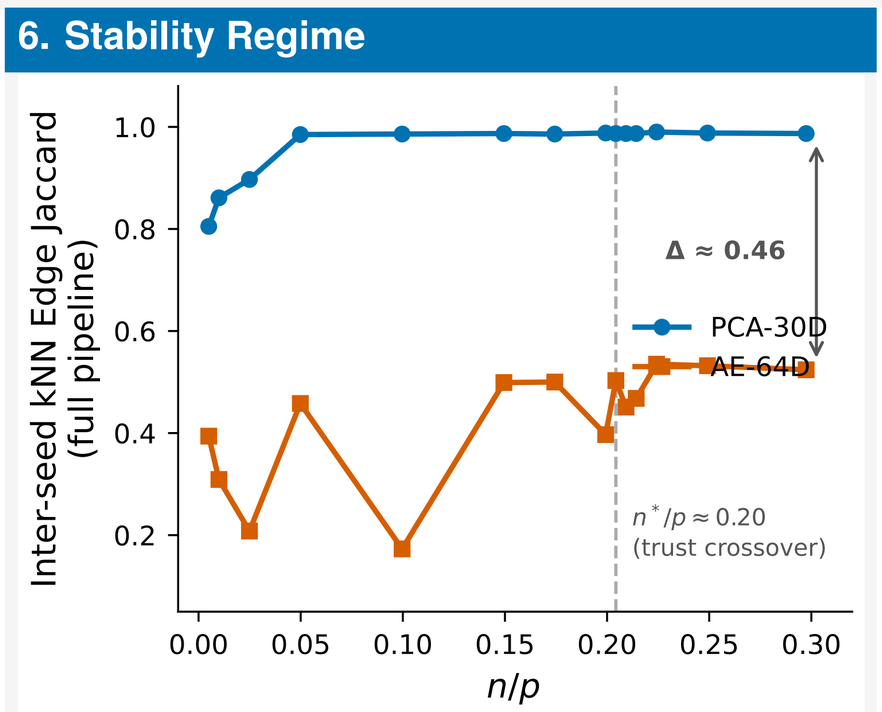


AE-64D yields $\leq$ +1 pp trust on Global panels but loses **8–10 pp** on LowDensity. Edge-Jaccard stability: PCA **0.88–0.91**; AE **0.16–0.52**.

Key Takeaways

1. **PCA geometry is robust:** SS $\geq$ 0.91 under 95% marker dropout.
2. **Monotonic kNN degradation:** no cliff effects; enables curator-side quality thresholds.
3. **Correspondence-free QA:** geometry diagnostics + robustness curves work without shared identifiers.
4. **AE not justified in $n \ll p$:** marginal trust gains vs. 2–5$\times$ worse topological stability.

6. Stability Regime



AE matches PCA trust at $n/p \gtrsim 0.20$ but the inter-seed stability gap persists ($\Delta \approx 0.46$): learned embeddings are not operationally justifiable when $n/p < 0.3$.

7. Failure Mode

In ultra-wide regimes ($n/p < 0.2$), learned embeddings exhibit high inter-seed variance, producing **non-reproducible kNN graphs** and unstable downstream community structure.

Indicator	$n/p < 0.2$
AE seed variance	high
Graph topology	unstable
Clustering reprod.	non-repr.

Stability, not trust, is the **binding operational constraint**.

References [1] Venna & Kaski (2006). *Neural Networks*. [2] McInnes *et al.* (2018). arXiv:1802.03426. [3] Kilian *et al.* (2012). *Methods Mol. Biol.* [4] CIP Dataverse: data.cipotato.org